01 / RESEARCH PROJECT

Prompt Depth and Regularization in MaPLe

A YZM5110 research project: reproducing MaPLe and testing a simple deep-prompt regularizer on EuroSAT.

THE QUESTION

Does using prompts in more layers always help with new classes?

THE BASELINE

MaPLe (CVPR 2023) adapts the visual and text branches of frozen CLIP together.

MY EXTENSION

I tested prompt depth and added DAPR: regularization for deep text prompts.

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02 / SELECTED PAPER

What was the MaPLe paper about?

MaPLe learns connected prompts for image and text processing, while the pretrained CLIP encoders remain frozen.

IMAGE BRANCH

An image enters CLIP. Visual prompts guide the image representation.

TEXT BRANCH

Class names enter CLIP. Text prompts are linked to visual prompts.

Main idea: adapt both modalities without full model fine-tuning

03 / PAPER COMPARISON

Related work in one table

The five papers reviewed in my report explain the path from CLIP to MaPLe.

PAPER	MAIN IDEA	LINK TO OUR STUDY
CLIP	Pretrains image and text encoders together.	Frozen starting model for all our experiments.
CoOp	Learns text prompts for a new task.	Shows that prompts can replace full fine-tuning.
CoCoOp	Creates text prompts based on each image.	Aims to improve transfer to unseen classes.
VPT	Adds learnable prompts inside a vision transformer.	Shows prompt learning on the image side.
MaPLe	Learns connected visual and text prompts.	Our baseline; we test depth and add DAPR.

04 / MY APPROACH

How did I approach the paper?

I started with a MaPLe reproduction, tested different depths, and then changed the training loss.

STEP 1 / REPRODUCE

Run MaPLe on EuroSAT using the published-style 16-shot setting.

STEP 2 / EXPLORE

Change depth K = 1, 3, 5, 7, 9 and compare base and novel accuracy.

STEP 3 / EXTEND

Add DAPR and test regularization strength and depth weights.

My contribution: controlled depth tests and a targeted loss change to MaPLe

05 / MY CONTRIBUTION

DAPR: a simple new loss term

I added an L2 penalty to MaPLe deep text prompts. CLIP remains frozen; no new trainable weights are added.

ORIGINAL MODEL

Frozen CLIP encoders

+

MAIN OBJECTIVE

Classification loss

+

ADDED OBJECTIVE

Deep-text prompt L2

$$L_{total} = L_{CE} + \lambda \cdot (1 / D) \sum w_i \|P_i\|^2_{mean}$$

- $\gamma = 0$: equal weights | $\gamma = 1$ or 2: deeper prompts receive larger weights

06 / MAIN EXPERIMENTS

What experiments did I conduct?

MaPLe with frozen CLIP ViT-B/16 · EuroSAT RGB · 16 shots · 5 epochs · three random seeds.

CLASSES

5 + 5

Base + novel

PROMPT DEPTH

1–9

K = 1, 3, 5, 7, 9

DAPR WEIGHTS

0 / 1 / 2

at K=9, λ=0.1

LOSS STRENGTH

3

0.01, 0.1, 0.5

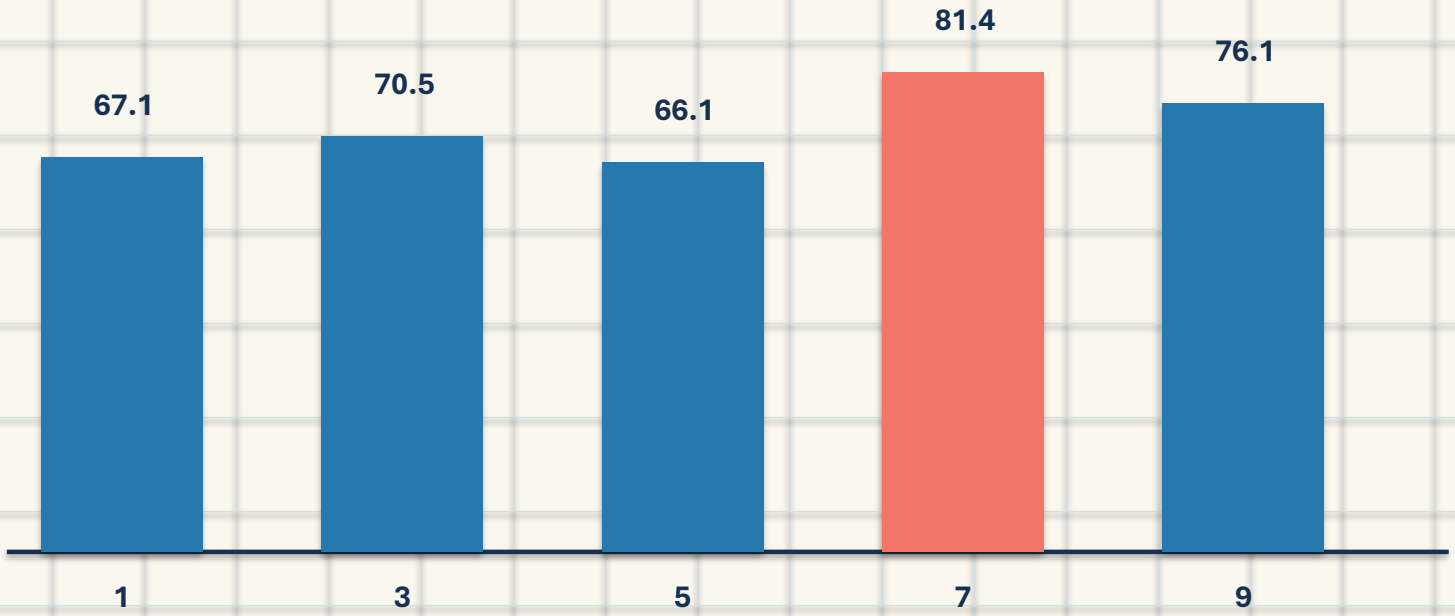
Main tests: five depths, three gamma settings, three lambda settings, then a K=7 check.

- Evaluation: accuracy on base and unseen classes + harmonic mean (HM)

07 / FIRST MAIN FINDING

More prompt depth was not always better

Mean harmonic mean (HM) from three seeds: K=7 performed better than the deeper K=9 baseline.



BEST BALANCE

81.37%

Baseline K=7

BEST BASE ACC.

93.58%

Baseline K=9

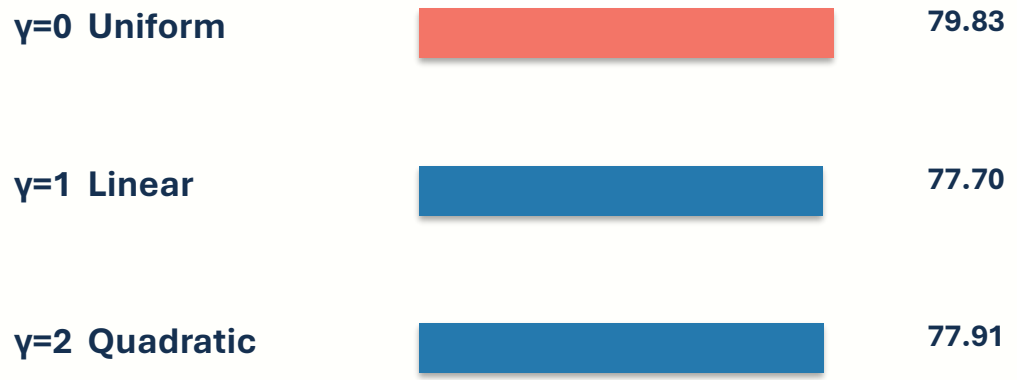
K=7 needs 22.2% fewer trainable parameters than K=9—and has a higher HM.

08 / REGULARIZATION TESTS

Which type of regularization worked?

At K=9, γ=0 gave the highest HM. Changing λ also changed performance strongly.

γ STUDY / K=9, λ=0.1



λ STUDY / K=9, γ=0



- Depth-weighted penalties (γ = 1 or 2) did not beat equal weights (γ = 0).

09 / FINAL RESULT

Best observed result: 82.13% HM

The final K=7 comparison shows a small increase in mean HM with uniform DAPR.

MAPLE · K=7

81.37%

Mean HM without penalty

DAPR · K=7

82.13%

γ=0 / λ=0.1

HM CHANGE

+0.77 pp

Exploratory result

At K=9: baseline 76.09% HM → uniform DAPR 79.83% HM (+3.74 pp).

- HM measures the balance of accuracy on seen classes and unseen classes.

10 / CONCLUSION

What did I learn from this project?

More depth did not guarantee better transfer. Regularization helped in some settings, but not every weighting scheme.

MAIN FINDING

Prompt depth has a non-monotonic relationship with generalization. K=7 is the best original baseline in this EuroSAT sweep.

MY RESULT

Uniform DAPR reached 82.13% HM at K=7, compared with 81.37% without it.

LIMITATIONS

Only EuroSAT and three seeds; test-guided choices. The improvement is not statistically confirmed.

Thank you · github.com/siyahkut/yzm5110_research_1